

LESSON GUIDE

Technology Plus

Unit 7: Security

6.1.4 – Privacy

Lesson Overview:

This lesson explores the importance of privacy in the digital age. Students will examine how social networking, email, file sharing, and instant messaging all present privacy challenges, particularly with personally identifiable information (PII). The lesson also highlights government regulations like GDPR and practices like cookie consent that exist to protect users.

Students will:

- Review examples of personally identifiable information (PII) and explain why protecting it is critical.
- Define privacy and recognize risks associated with different communication and data-sharing platforms.
- Examine government regulations and how they aim to protect users and their data.

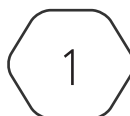
Guiding Question: What are some concerns involving privacy and how can you protect your information in an interconnected digital world?

Suggested Grade Levels: 8-12

Technology Needed: No

Approximate Time Required: 60-75 minutes

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Lesson Background

Standards:

CompTIA Tech+ FC0-U71 Objective

- 6.1 – Explain fundamental security concepts and frameworks.
 - Confidentiality, integrity, and availability
 - **Privacy**
 - **Social networking sites**
 - **Email**
 - **File sharing**
 - **Instant messaging**
 - **Personally identifiable information (PII)**
 - **Government regulations (e.g., General Data Protection Regulations [GDPR])**
 - **Cookie consent**
- 6.2 – Explain methods to secure devices and security best practices.
 - **Privacy considerations**
 - **Social networking sites**
 - **Email**
 - **File Sharing**
 - **Instant messaging**
 - **AI**

Lesson Background

Background Information:

Teacher Note: Although the main testing objectives listed are distinct, being aligned to 6.1 and 6.2, there is a correlation between the subobjective's software tools/platforms mentioned. Due to this connection, the objectives were combined in this lesson to offer the fundamental concepts of each, associated security risks, and security best practices for a more cohesive lesson.

Privacy goes beyond security by addressing how personal information is collected, used, and shared. Students should understand that personally identifiable information (PII) includes anything that could identify a person including full name, SSN, IP address, email, biometric data, and more.

There are risks involving privacy that are associated with various communication applications or tools including:

- Social networking sites: Oversharing, weak privacy settings, phishing links.
- Email: Spam, phishing, malware attachments.
- File sharing: Unsecured platforms risk data leaks or malware.
- Instant messaging: Encryption varies; risk of impersonation or leaks.

There are also regulations that aim to hold organizations accountable for the privacy of its users. The General Data Protection Regulations (GDPR) in the EU is one of the strictest frameworks. It requires informed consent, data minimization, and user rights to access/delete data. Other regulations that are required in most countries now require cookie consent banners are part of compliance to notify the user that sites can and often do save data even if they are just browsing the site.

Materials Needed:

Materials per Class	• Lesson Guide 6.1.4 – Privacy • Lesson Slides 6.1.4 – Privacy
Materials per Student	• Student Handout 6.1.4 – Privacy • Student Notes 6.1.4 – Privacy • Optional Assessment 6.1.4 – Privacy

Lesson Background

Cyber Terms:

Term	Definition
Personally identifiable information (PII)	data that could reveal a person's identity
privacy	the principle that we should have the ability to control how personal information is used, stored, and collected
end-to-end encryption	process when a message is encrypted, or converted from readable to unreadable text, on the sender's device, transmitted, and decrypted on the receiver's end back to readable text
General Data Protection Regulation (GDPR)	states that European citizens, regardless of where a company is based, have full rights and privileges to how their data is collected, used, and managed
Health Insurance Portability and Accountability Act (HIPAA)	protects personal health information from being disclosed without the patient's information
Payment Card Industry (PCI)	regulates credit cards and other financial data for users' privacy
first-party cookies	pieces of data collected by websites that are limited to that specific site and can be used to make navigating the site more seamless
third-party cookies	pieces of data collected by websites that function across multiple sites, often for advertisements or tracking

6.1.4 – Privacy

Guiding Question: What are some concerns involving privacy and how can you protect your information in an interconnected digital world?

Lesson Launch (3-5 minutes)

- Start by presenting students with the image of a social media post by the high school student:
 - Ask what risks are noticed in the post related to oversharing information and if students have done something similar.
- You can click to show specific cases of oversharing information including:
 - Full name listed as the username.
 - School name.
 - Paired with the post date, the exact birthday could be determined.
 - The grade level could be used to determine age.
 - Location including hometown, school, and when paired with other information a daily schedule could be determined.

PII and Privacy (5-10 minutes)

- Distribute Student Handout 6.1.4 – Privacy to students and encourage them to fill in answers as you go through today's lesson.

Personally Identifiable Information

- Personally identifiable information (PII) is data that could reveal a person's identity.
- Examples of PII include a person's name, birthday, phone number, address or more. Information that could cause substantial issues with privacy are considered sensitive personally identifiable information (SPII) and include data like a driver's license, Social Security number, account information, or other records containing in depth data on a user.

Privacy

- **Privacy** is the principle that we should have the ability to control how personal information is used, stored, and collected.
- As more transactions and data move to being online, more personal information is readily accessible, and privacy becomes more difficult to maintain.

Keeping Privacy Private (25-30 minutes)

Social Media

- Although social media can be useful with staying connected to others, aid with growth for businesses, and they allow you to set security settings, they still pose a huge risk to privacy. The main risk is that those sites own your personal data per the terms you agreed upon when you created the account.
 - This means they can sell that data to other companies, mostly for advertising.
- The data they can collect and then own them includes images, videos, location tracking, times spent on the site, and more. Some of this is collected in the background without the user actively being in the application or on the website.
- Protecting data from social media websites includes:
 - Adjusting privacy settings to control who can see your information by setting accounts to private, limit visibility to only friends/family, removing tracking if possible.
 - Reviewing followers and limit who has access to view your account and posts.
 - Sharing carefully and avoid posting anything that could be considered sensitive or that could be used be releasing too much private information.
 - Watching the add-ons as many social media sites will connect to games or third-party apps outside of the platform. These have their own permissions that might reveal PII without you realizing it.

Email

- As one of the earliest ways to communicate across the internet, email has risks associated with it that have always existed. Network security has improved and when used correctly prevents email from being intercepted and read during transit; however, the email itself is not protected on the user's end or once it is on an email server.
- **End-to-end encryption** means the message is encrypted, or converted from readable to unreadable text, on the sender's device, transmitted, and decrypted on the receiver's end to be read. This prevents even the email provider from viewing the contents of the email. This option usually requires the user to select it as an option prior to sending an email.
- Other privacy concerns include who owns the email account, i.e. personal vs. work/school. Using an email through an organization typically means they have full access to that email account, meaning none of the messages are private.
- It is also important to note that emails can be subpoenaed by courts, so considering what is emailed is also important.

Instant Messaging (IM)

- Instant messaging (IM) is like a union between social media and emailing, pulling in some benefits while also keeping some of the risks.
- IM platforms function like social media, often owned by a parent company that runs a social media site already, meaning they can potentially collect a large amount of data on users. That data can be sold to third parties, often for advertising purposes. The messages themselves are not necessarily owned by the platform; however, by agreeing to their terms, users are allowing them to have control over them and how they are managed. This is often the case with anything sent online. Once posted,

uploaded, sent, etc., **you have no control over what the recipients do with it.** This is true for any form of online communication.

- Keeping instant messages private requires similar techniques as email.
 - Use end-to-end encryption when available, some provide it by default, which is nice to have.
 - Be mindful of what is always shared.
 - Note that some IM services are integrated into software tools utilized by organizations such as Microsoft Teams or Google Chat, so like email, be mindful of what is sent as the organization you are affiliated with may be able to read it.

File Sharing

- Old methods of file sharing, such as the File Transfer Protocol (FTP) were not secure by any means and even included login credentials when used.
- Newer services are more secure but still have the risk of having files accessible by unauthorized means from improper sharing settings, using unsecured networks, and poor security practices with the files itself.
- File sharing can be secured and remain private by:
 - Ensuring the account is properly secured using strong credentials and multi-factor authentication if available.
 - Use only secure networks to access shared files.
 - Review the permissions of the file to check who can view, edit, or manage the file.
 - Consider using expiration dates for shared files.
 - Share a copy of the original file and not the original.

Artificial Intelligence (AI)

- AI has been around to some degree for a while, but with advancements and integrations into existing programs and tools the associated risks have grown.
- AI platforms vary drastically in how they function and are used. Generative AI tends to save chats, user data, and other inputs. Most of this is saved to help build and develop new AI models. Based on the type of AI platform or account used, personal information may also be saved, such as payment data and contact information.
- A positive note is that most larger businesses that have paid models for use do not share or sell user data as they are incorporated in ways that users are possibly implementing it in proprietary ways or with their already existing products. In either case, policies do change, and it is important to read and keep up to date with them.
- Other ways to protect privacy while using AI includes:
 - Never using or adding PII or sensitive information with AI tools. This includes photos of yourself and should there ever be a leak, that image could be used in malicious ways, especially as facial recognition imitation becomes more prevalent.
 - Be aware of what data is tracked by apps that are meant to use AI to develop trends. Health and finance apps are doing this more, possibly exposing that data to other things or people.

- Be cautious of AI tools that are in beta or experimental forms. There is no guarantee that they are fully secure and may have vulnerabilities that could release your personal information.

Regulating Privacy (10-15 minutes)

General Data Protection Regulation (GDPR)

- The **General Data Protection Regulation (GDPR)** in the European Union is widely known as one of, if not the strongest user privacy law in effect. Citizens living in countries within the European Union, regardless of where a company is based, have full rights and privileges to how their data is collected, used, and managed. It also allows them to change their preferences or request that it be deleted. Failure to do so results in heavy fines for the hosting company.
 - It is so strict that many companies now budget specifically for the GDPR because there is always a risk that data could be released through negligence, accidental, or malicious means. A U.S. based company that oversees several social media platforms, applications, and software was fined over a billion dollars in one year due to the GDPR; however, even with amounts such as this, it is often just a drop in the bucket for how much companies bring in on advertising from user data.
- At the international level, the **Payment Card Industry (PCI)** works to protect and regulate user data for credit and financial transactions and was created by the very cards people use. Think Mastercard, Visa, American Express, etc., as none are tied to one specific bank or company. Instead, they are independent corporations who have independent networks for financial transactions, and they enforce the policies of the PCI council. As with the GDPR, businesses do not want to violate anything related to the PCI regulations such as having Mastercard or Visa revoking the ability to use them could result in huge setbacks with a business.
- In the U.S. regulations are created to focus more on specific types of data. An example is the **Health Insurance Portability and Accountability Act (HIPAA)** protects personal health information from being disclosed without the patient's approval. Some states have also moved to create their own laws and regulations to protect user data.

Cookie Control

- Cookies are small files downloaded from sites and tied to your browser that save pieces of user data.
 - **First-party cookies** are limited to that specific site and can be used to make navigating the site more seamless.
 - **Third-party cookies** are used across multiple sites and mainly related to online ads.
- Sites now alert you to what cookies are being used and you must click to accept them. Many browsers, such as Firefox and Opera, can block third-party cookies by default now upon setup. Some do not, so it is important to review policies.
- Both cookie types are now scrutinized and are under more regulations with only necessary first-party cookies that are critical to the site's functionality being allowed automatically.

Putting It All Together (3-5 minutes)

- What are some examples of PII, and why is it critical to protect it?

Information that gives away a person's identity including but not limited to name, birthday, phone number, address, driver's license, social security number, account information, and more. It is critical to protect as it could be used maliciously as with identity theft, fraud, theft, or other malicious means.

- Which communication tool/app discussed today do you think poses the greatest risk for users? Why?

Answers will vary. Often social media is considered the greatest risk because people overshare, there are vulnerabilities, or poorly implemented security settings.

- What is your opinion on government intervention in terms of regulating what companies can do with your data?

Answers will vary but should provide an active discussion. Many people are against heavy regulations because it limits the freedom of the internet but also demand protection when things go awry. It is finding a happy medium that is difficult.